TASK REGISTER · REV 2026-08-08

Task register — `final-dataset`

Written 2026-08-07. The `#NN` ids are referenced throughout `IMPLEMENTATION-PLAN.md`, `BUILD-DEPENDENCY-GRAPH.md` and `HANDOFF-FINAL-DATASET.md`. They previously lived **only in a session task tool**, so a fresh agent could read `#21` in three documents and find no definition anywhere. This file is the definition. **It is the crosswalk between all three numbering schemes** — task id, graph node, and Phase 0 item.

Nothing here has been submitted to AWS. Every job needs a platform submission and a human release.

Wave 0 — the critical-path code

#	task	graph	Phase 0	changes the plan?	est
#28	**Split the BIG bundles — the ROUTE matters more than the task.** `--shard/--of` strides *bundles*, so DCLM's 410B is one child at **10.85 h even on a whole 32-vCPU instance**; FineWeb-Edu's 252B is 6.67 h. **Try the synthetic** `domain_column` **fan-out first (~2 h → path 7.75 h); the 12 h ordinal-range route buys only 0.64 h over doing nothing** (break-even ~11.5 h)	**C3b**	**3b**	**YES**	**~2 h**, or 1–2 d
#22	**Replace the dedup set with a flat** `np.uint64` **pre-pass.** The `set` needs **27.92 GB** for DCLM in a 15.03 GB container (§5.2a)	A2a + A2b	4	no	1 d
#21	**Wire the FinePhrase id partition** into `_reader_for`. Cannot be retrofitted after tokenization. **Ships with the budget correction or not at all**	C1	1	**YES**	~5 lines
#9	**Decide** `SHARD_TOKENS`**.** Code says 25,001,984; the report now agrees. Confirm and stop carrying two values	B6	11	**YES**	1 h
#24	**Rebuild the decon index from raw benchmark fields**, not 5-shot renders. Gates `FREEZE`	B5	12	no	4 h

Wave 0 — parallel, distinct files, no coordination

#	task	graph	Phase 0	est
#23	**Pin** `tokenizers` in `pyproject.toml`. It is imported at `corpus_build.py:631` and declared nowhere	B1	5	1 line
#10	**Thread the Gate A profile checks + raise** `max_pool_connections`. ~100 lines, not the ~20 an earlier draft said. **Ship this BEFORE #29** — deleting `verify --deep` exposes `GA1`, and unthreaded that is 5.08 h, which makes #29 worth exactly nothing	B3	6	1 d
#29	**Declare** `ChecksumSHA256` **on the shard upload, then retire** `verify --deep`. `corpus_build.py:463` calls a plain `put` and computes the digest one line too late, so S3 never verifies it. A verified PUT is rejected server-side on mismatch (proven live: `BadDigest`, and **no object is created**), and `CopyObject` recomputes on the publish hop — so the 1.49 h re-hash is redundant **once the checksum is declared.** Keep the code for VENDORED corpora	**B7**	**13**	~5 lines
#16	**Drop** `data_provenance_initiative` — ships GSM8K in Flan CoT format at 6 repeats. 0.51% of tokens	B4	9	registry edit
—	**Fix the boundary-marker prefix guard** in `corpus_pack.py`, and the test asserting the table length is 1	B2	8	~5 lines
—	**Record** `FilterStats` **in the receipt.** Until then no dedup claim is auditable	—	7	~10 lines
#11	**Query the live validator timeout.** `edullm-validator:12`'s is recorded nowhere. Read-only	—	10	1 call

Wave 0 — measurements

#	task	graph	est
#26	**Measure in-region S3 + HF CDN bandwidth. RUN THIS FIRST** — every read estimate borrows ~85 MB/s from an S3 measurement, and one reconciliation implies the CDN is ~8.4 MB/s	M1	10 min
#14	**Dolma 3 adult-content prevalence**, sampled at **random offsets** — a prior attempt could not separate signal from HuggingFace preview ordering. **Blocking for that source**	M2	1 h
—	~~Nemotron-CC-Math's real dolma2 count~~ ✅ **DONE 2026-08-07** — **134.0B** (472,213,218,716 bytes × 0.283686 tok/byte, 1,920 random-offset docs, seed 42; `3` ≈ 83.6B + `4plus` ≈ 50.4B). Measured by a teammate with gate access. **This closed the M3 node, NOT task #17**	M3	done
#17	**Re-run Marin's 13-gram contamination scan on Nemotron-CC-Math.** A different task from the count above, and **still open.** The original scan found **11,868 contaminated documents against MATH500 alone — 13.2× the publisher's entire reported removal budget** — including verbatim GSM8K *test* items at Jaccard 1.0. **Now unblocked:** the gate is accepted and 107+ GiB is already staged in S3, so this runs in-region against staged bytes	—	1 job
—	**Mean doc length for 5 unmeasured stage-2 sources.** The dolma3 QA source is the one plausibly near the 20-token EOS floor	M4	1 h

Deferrable — do NOT put in the first image

#	task	graph	why deferred
#25	**File-shard the VAL bundles.** 43% of bytes moved serves 0.39% of tokens. **NOT the same item as #28** — this one is a read-volume saving, #28 is a wall-clock prerequisite	C3	∞ slack; contends with #21 on `_reader_for`
—	**Wire** `bytes_fetched` through from `corpus_read` to the build	C11	instrumentation only

Then, in order

#	task	note
#20	**Freeze the full two-stage plan.** THE REAL GATE. Its duration is a decision, not a computation	after every "changes the plan" item lands
#6	**Ingest.** Stage → smoke test → build in waves → publish two datasets → Gate A → `verify --deep` → promote	
#1	Register `reservoir-dolma2` with the platform	independent of this corpus

Downstream of the corpus — training-side, not build-side

#	task
#19	**Fix** `MoELoadBalancingLossGranularity` before any training run. Cannot be annealed in — switching at 10% recovers ~55%
#7	Run probe-fitted mixture selection before the real run
#18	Use BPB, not accuracy, for any ablation at this scale — MCF is random below ~400B tokens
#8	Store MTLD as a per-document label **at ingest** — labels are inside `manifest_sha256`, so unbackfillable
#13	Settle SuperBPE empirically with the existing byte-matched A/B corpora (~$600)
#12	Fix the missing EOS in the SuperBPE tokenizer. **Affects two already-published corpora, NOT this one** — dolma2's `eos_token_id` is 100257 and its check runs

Closed

#	task	outcome
#5	Measure the real FinePhrase overlap	**0.2683 distinct** on a complete-column read of 287,000 ids
#15	Interleave sources within shards	**Not expressible in data.** `labels` is one dict per shard path and Gate A compares by full dict equality. Trainer-side (#19)
#4	Wire `keeps_id` into the build path	superseded by #21, which is the same fix scoped to the reader
#27	Fix 23 verified doc inconsistencies	this pass